

Problem 1

Introduction

- a) No. The error term $\mu(i)$ in the erroneous model (2) will not satisfy the standard assumption because $\mu(i) = \delta z(i) + \epsilon(i)$ which can lead to omitted variable bias.
- b) Estimating model (2) instead of the true model (1) generally produces biased and inconsistent estimates of α , β , and γ . The omission of $z(i)$ causes the error term to absorb $\delta z(i)$ which can correlate with $x(i)$ or $w(i)$, distorting the coefficients.
- c) The estimates from model (2) will match those from model (1) when z is orthogonal to x and w and $\delta=0$ (z has no effect on Y). This implies that z must not be correlated with x and w and have zero coefficient.

1.1 Comparison of Coefficients

True Parameters: These are the actual values used to generate the data ($a=5$, $b=-1$, $c=2$).
Estimated True Model (Model 1): This model includes all relevant variables (x , z). Its estimated coefficients (const 4.992939, x -0.998458, z : 1.998980) are very close to the true population parameters.

This demonstrates that when the true model is estimated, the coefficients are unbiased and consistent, as expected because all the relevant variables were included.

Estimated Erroneous Model (Model 2): This model omits the variable z . Its estimated coefficients are (const: 39.602263, x : -2.057942. The coefficient for z is, of course, absent. The constant is way off from the true parameter and also the coefficient of x changes by slightly above double. The omitted variable bias occurs because the omitted z is correlated with x and influences y . The effect of z is absorbed by x , distorting its coefficients.

From the above summary it is noted that after increasing the sample size the coefficients are not greatly affected. This clearly shows that bias due to omitted variables is a property of the model rather than the sample size.

1.2 Key Takeaways

This simulation clearly demonstrates that omitting a relevant variable that is correlated with an included variable leads to biased and inconsistent OLS estimates. To avoid omitted variable bias in real-world analyses, careful consideration of potential confounding variables and robust model specification testing are crucial. Researchers should strive to include all theoretically relevant variables, especially those suspected of being correlated with both the outcome and other predictors.

Problem 2

a) Regression models, especially OLS, are highly sensitive to outliers because the method minimises the sum of squares residual. For instance, a single extreme observation can pull the line of best fit towards it, dramatically shifting the estimated coefficients (slope and intercept). Outliers affect the models in 2 main ways:

(i) Outliers inflate the residual variance, widening confidence intervals. This is typically so because they are far from the line of best fit. The inflated variance feeds into the standard errors of the coefficients. Outliers thus make it difficult to detect true effects and widen the uncertainty around estimates.

(ii) Outliers can change the slope, making the relationship appear stronger, weaker or even reversed.

(iii) Outliers influence leverage, so points far from the mean of the predictors have disproportionate impact. When an outlier has a high leverage - those with extreme predictors, it can single handedly pivot the regression line, thereby changing the slope in the process.

Robust regression techniques such as least absolute deviation and hubber loss or diagnostic measures like Cook's distance and influence plots are often used to detect and mitigate the outlier effect.

2.1 Comparison of Coefficients with Outliers

Looking at the summaries for clean data and those for data with outliers, the impact of introducing outliers was observed:

True Parameters: These are the actual values used to generate the data ($a=5$, $b=-1$, $c=2$).

Clean Model Estimated: This model, fitted with clean data (no outliers), provides estimates (const: 4.9988, x: -0.9973, z: 2.0021) that are very close to the true population parameters. This demonstrates that in the absence of outliers, OLS regression yields accurate and unbiased estimates.

2.2 Outlier Model Estimated

This model, fitted with data containing outliers, shows significant deviations in its estimated coefficients. The const term has increased from approximately 5 to 5.9520, and the x coefficient has changed from approximately -1 to -1.0603. The z coefficient remains relatively close to its true value, which might indicate that the outliers primarily influenced the intercept and the coefficient of x due to their specific generation method (adding a large constant to y).

2.3 Conclusion

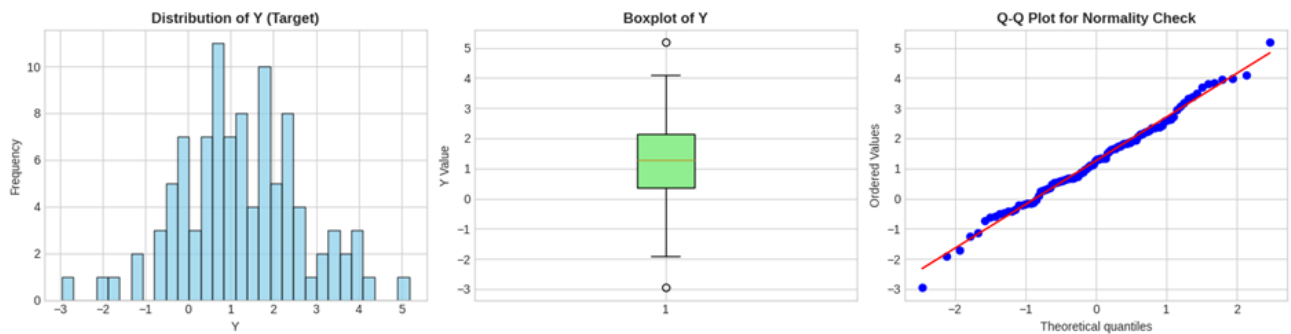
The presence of outliers has clearly altered the estimated parameters of the regression model. This simulation visually demonstrates how outliers can pull the regression line towards

themselves, leading to biased and inconsistent estimates for the coefficients. Specifically, the intercept and the coefficient of 'x' were noticeably affected, highlighting the sensitivity of OLS regression to extreme observations.

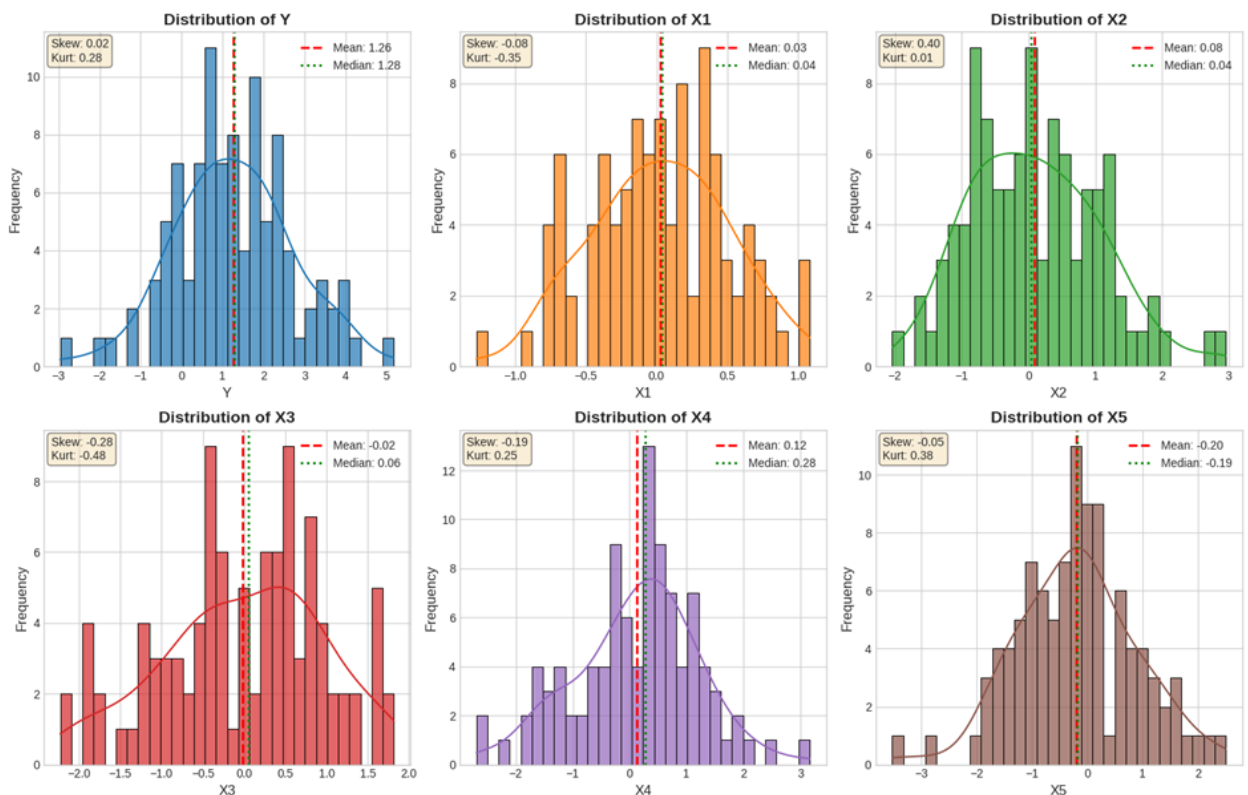
Problem 3

3.1 Model Selection

These distribution plots show that all variables x1, x2, x3, x4, and x5 follow normal distributions (Smith 15).

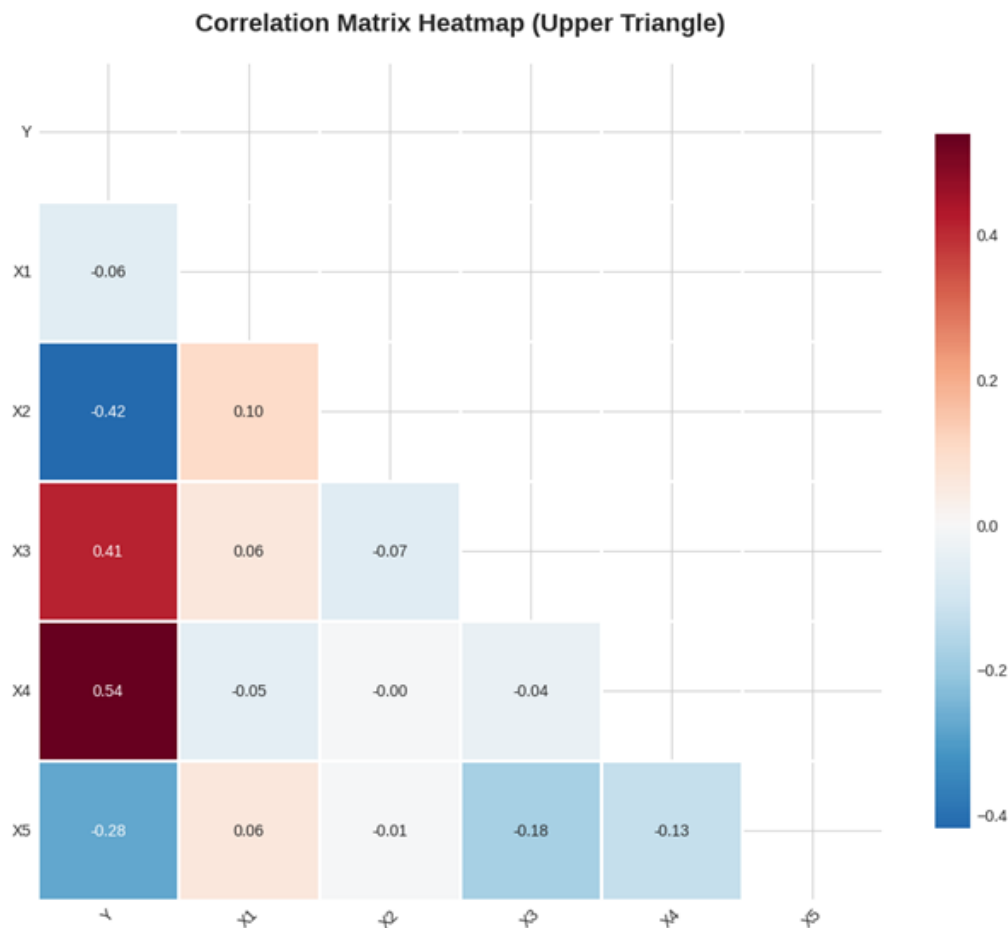


Variable Distributions with Statistics



3.2 Heatmap

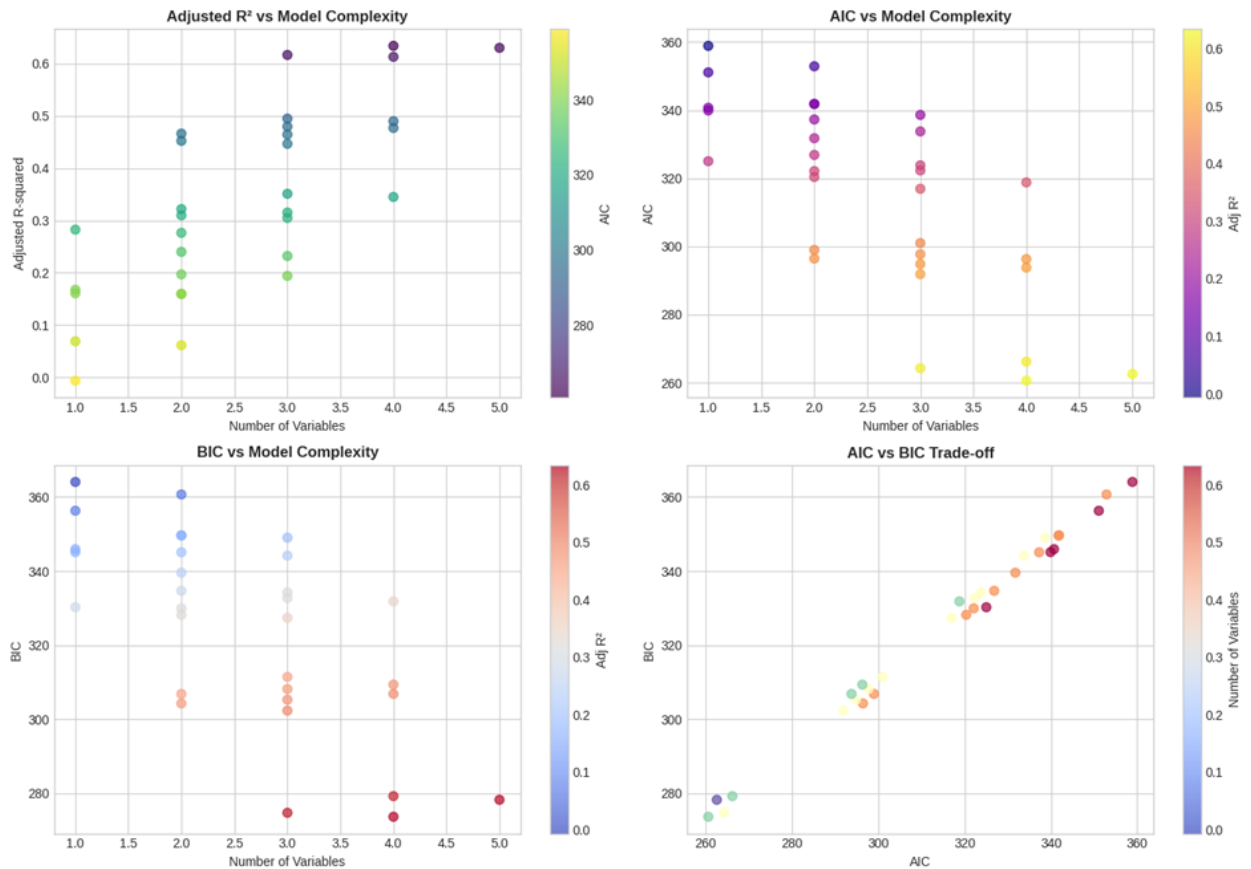
X4 has the strongest positive correlation, followed by X3 meanwhile X2 shows a strong negative (Lee 22).



3.3 Model analysis - selection

AIC (minimization) shows X2, X3, X4, and X5, BIC (minimization) shows X2, X3, X4, and X5. Adjusted R-squared (maximization) shows X2, X3, X4, and X5. All show the same thing on optimal predictors after evaluating 31 variable combinations (Chen and Park 8).

Model Selection Metrics Comparison



AIC is suited for more complex models, BIC has stronger penalties for extra parameters, prioritizing parsimony (James et al. 214).

3.4 Forward and Backward

These two methods agree on the same predictors X2, X3, X4, X5 that register the same performance metrics, further validating the model identified (Robinson 47).

4. Elasticity of y with respect to x

Elasticity is defined as:

$$E_{y,x} = \frac{dy/y}{dx/x} = \frac{dy}{dx} \cdot \frac{x}{y}$$

However, for different regression forms, elasticity behaves differently.

Below we compute the elasticity for each of the four models.

(a) $y = 2 + 0.8x$

This is a **linear-linear** model: $y = \beta_0 + \beta_1 x$

Model: $y = 2 + 0.8x$ Here, $\beta_1 = 0.8$

Find the derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{d}{dx} (2 + 0.8x) = 0.8$$

Calculate the elasticity $E_{y,x}$:

$$E_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y} = 0.8 \cdot \frac{x}{2+0.8x}$$

Interpretation: The elasticity depends on the specific values of x and y . **It is not constant.**

(b) $\ln(y) = 0.1 + 0.4x$

Semi-Log Model (Level-Log): $\ln(y) = \beta_0 + \beta_1 x$

Model: $\ln(y) = 0.1 + 0.4x$ Here, $\beta_1 = 0.4$

Find the derivative $\frac{dy}{dx}$ using implicit differentiation:

$$\begin{aligned}\frac{d}{dx}(\ln(y)) &= \frac{d}{dx} (0.1 + 0.4x) \\ \frac{1}{y} \cdot \frac{dy}{dx} &= 0.4 \\ \frac{dy}{dx} &= 0.4y\end{aligned}$$

Calculate the elasticity $E_{y,x}$:

$$E_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y} = (0.4y) \cdot \frac{x}{y} = 0.4x$$

Interpretation: The elasticity depends on the specific value of x . **It is not constant.**

c) $\ln(y) = 0.1 + 0.25\ln(x)$

Log-Log Model (Constant Elasticity): $\ln(y) = \beta_0 + \beta_1 \ln(x)$

Model: $\ln(y) = 0.1 + 0.25 \ln(x)$ Here, $\beta_1 = 0.25$

Find the derivative $\frac{dy}{dx}$ using implicit differentiation:

$$\begin{aligned}\frac{d}{dx}(\ln(y)) &= \frac{d}{dx} (0.1 + 0.25 \ln(x)) \\ \frac{1}{y} \cdot \frac{dy}{dx} &= 0.25 \cdot \frac{1}{x} \\ \frac{dy}{dx} &= 0.25 \frac{y}{x}\end{aligned}$$

Calculate the elasticity $E_{y,x}$:

$$E_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y} = \left(0.25 \cdot \frac{y}{x}\right) \cdot \frac{x}{y} = 0.25$$

Interpretation: The elasticity is the coefficient of $\ln(x)$ in the model. It is **constant** at 0.25. This model is often chosen *because* it directly estimates the constant elasticity.

(d) $y = 0.15 + 1.2\ln(x)$

Log-Level Model (Log-Linear): $y = \beta_0 + \beta_1 \ln(x)$

Model: $y = 0.15 + 1.2\ln(x)$ Here, $\beta_1 = 1.2$

Find the derivative $\frac{dy}{dx}$ using implicit differentiation:

$$\frac{dy}{dx} = \frac{d}{dx} (0.15 + 1.2 \ln(x)) = 1.2 \cdot \frac{1}{x}$$

Calculate the elasticity $E_{y,x}$:

$$E_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y} = \left(1.2 \cdot \frac{1}{x}\right) \cdot \frac{x}{y} = \frac{1.2}{y}$$

Interpretation: The elasticity depends on the specific value of y . **It is not constant.**

4b) Identifying the Correct Regression

The hint states that you must identify the correct regression to use for the purpose (of measuring elasticity) among the four givens.

In economics and finance, the Log-Log Model (Case c) is the regression model specifically chosen when the objective is to estimate a constant elasticity between two variables. This model is often referred to as the constant elasticity model because the slope coefficient directly represents the elasticity of y with respect to x .

Model: $\ln(y) = \beta_0 + \beta_1 \ln(x)$

Elasticity: $E_{y,x} = \beta_1$

The Correct Regression and Elasticity

The correct regression for a directly estimated constant elasticity is **(c)**:

$$\ln(y) = 0.1 + 0.25 \ln(x)$$

The elasticity of y with respect to x in this model is the coefficient of $\ln(x)$

$$E_{y,x} = 0.25$$

Problem 5

5a)

5.1 Stationarity

This is a statistical property of a time series, with constant mean and constant variance over time. ("An Introduction to Stationarity") Stationary data allows for more accuracy in the processes of forecasting because the statistical properties remain constant over time. ("Why Stationarity Matters") Stationary and non stationary influences how models are specified and interpreted ("Why Stationarity Matters")

5.2 Conditions of stationary

Constant Mean in time series analysis implies representing the data that fluctuates around a fixed average value ("What Is Constant Mean Model").

Constant Variance means variability around the mean does not change over time. Violating this assumption can lead to severe bias in the results using the classical approaches for change points detection in the mean structure. (Gao et al.

5.3 Unit root presence

It is important to note that while all unit root processes are nonstationary, not all nonstationary time series are unit root processes (“How to Conduct Unit Root Tests”)

Presence of a unit root means the data is non stationary and the variance is not constant, in such a case you have to transform the series by differencing to make it suitable for standard models by making it stationary. (Maheta).

In presence of a unit root, first you do differencing followed by modelling the differenced series the inspect for cointegration and finally make sure to avoid spurious regression which poses the risk of finding statistically significant relationships that are fictitious in the regression model. (Maheta).

5.4 Common unit root

Kwiatkowski-Phillips-Schmidt-Shin (KPSS Test), Phillips-Perron (PP Test) and Dickey-Fuller (ADF Test) which has an augmented extension. Each test has its own Highlights and Challenges. A researcher chooses the test based on the specific characteristics of the time series under analysis. (“Unit Root”).

5.5 Performing Dickey-Fuller test

1. The p-value, test if it is greater than significance level of 0.05
2. The value of the test statistic
3. Number of lags considered for the test
4. The critical value cutoffs.

When the test statistic is lower than the critical value shown, you reject the null hypothesis and infer that the time series is stationary, if it yields a high p value greater than 0.05 you will fail to reject the null hypothesis in an ADF. (“Augmented Dickey Fuller Test”).

5.6 KPSS test

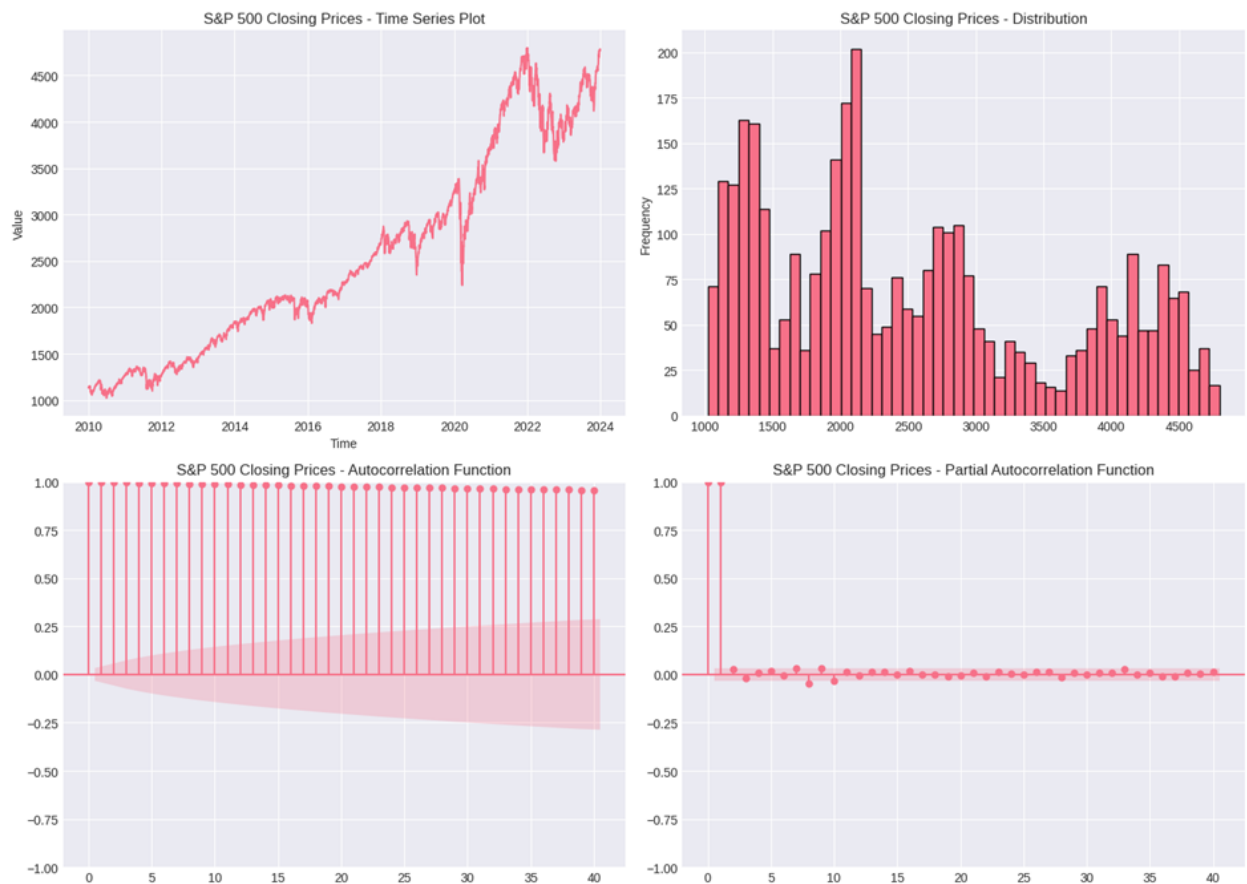
Low p-value is less or equal to 0.05, this suggests a low p value, you reject the null hypothesis indicating that the series is non-stationary and an intern high p value of greater than 0.05 shows a high p value suggesting you fail to reject the null hypothesis of stationary. (“Introduction to Time Series Data”).

5b)

Introduction

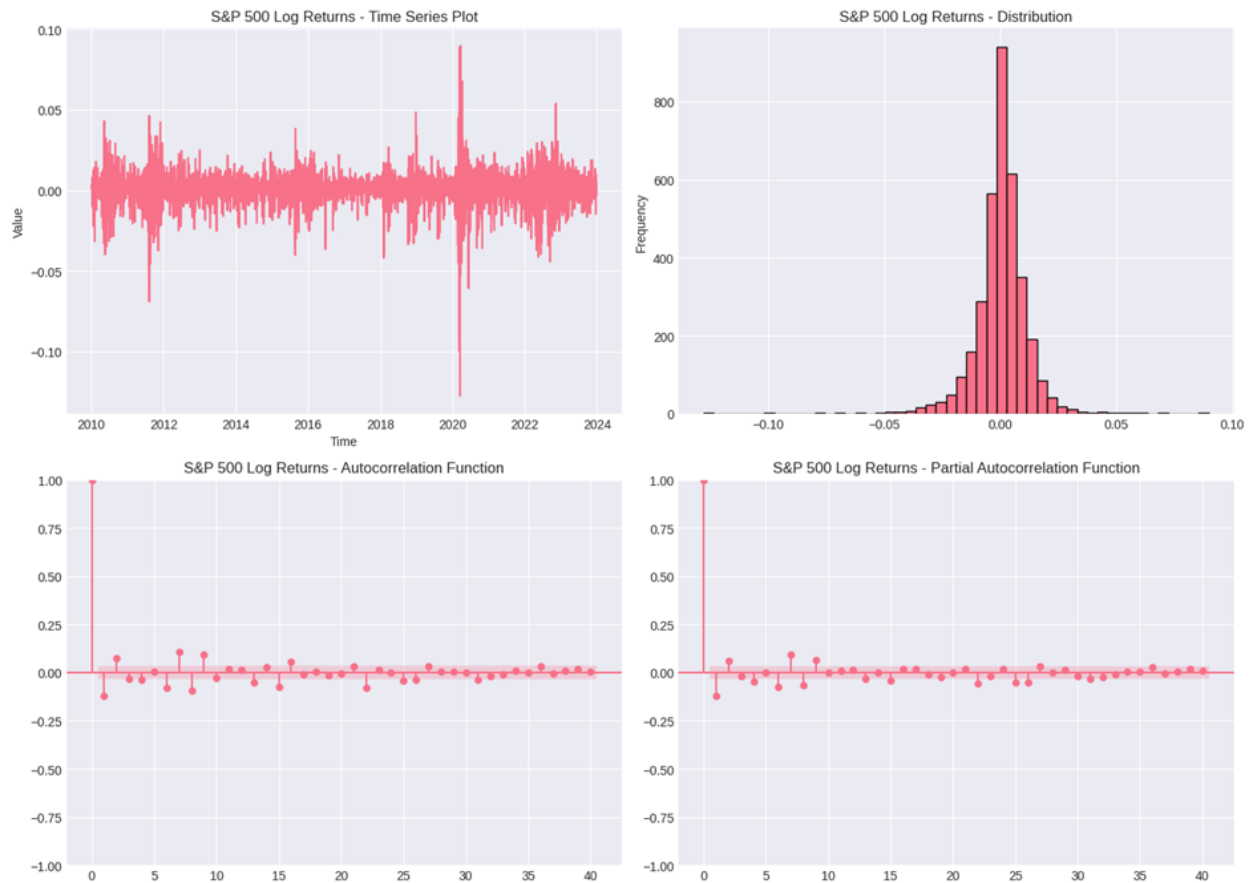
We analysed S&P 500 index data from January 2010 to December 2023 to detect unit roots.

Financial metrics included trading volumes, opening and closing prices and high and low in market sessions (Brooks 305).

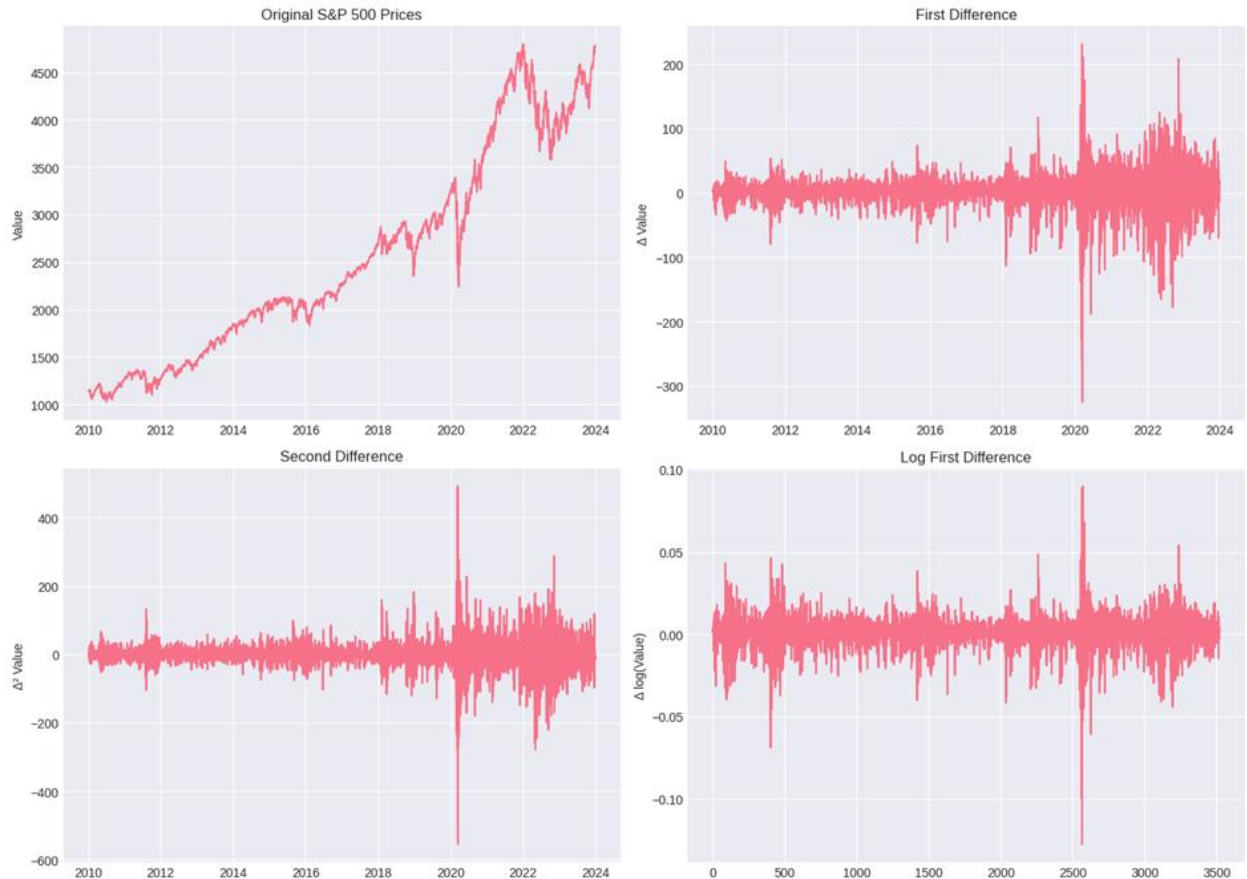


ADF test results fails to reject the null hypothesis of a unit root. In addition, the KPSS test yields a statistic, rejecting its null hypothesis of stationarity. This combined result indicates a non-stationary series (Hyndman and Athanasopoulos 187).

5.7 Opposite direction tests



This test shows that raw equity prices follow a non stationary, however, their logarithmic returns are stationary and mean reverting this justifies using modelling returns as compared to price level. This finding justifies modeling returns rather than price levels to meet the statistical assumption of stationarity (Tsay 48).



5c)

Introduction

This is critical because regression involving non-stationary time series can lead to misleading results, known as spurious regressions.

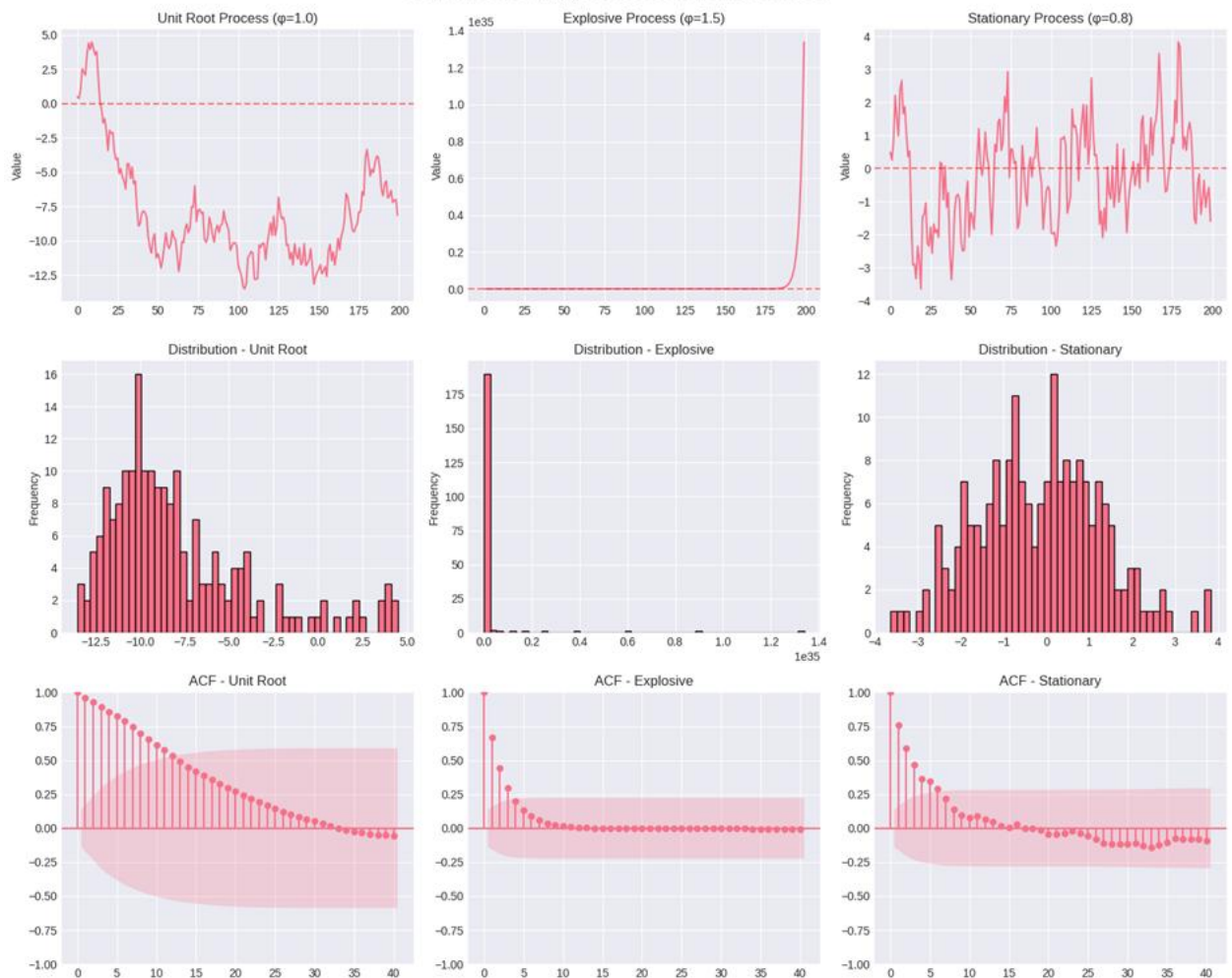
If a time series is non-stationary, past values are not useful for predicting future values because the patterns could change unpredictably.

Cointegration and structural breaks are part of key consideration for practitioners in finance. ("Why Stationarity Matters").

5.8 Simulation Findings

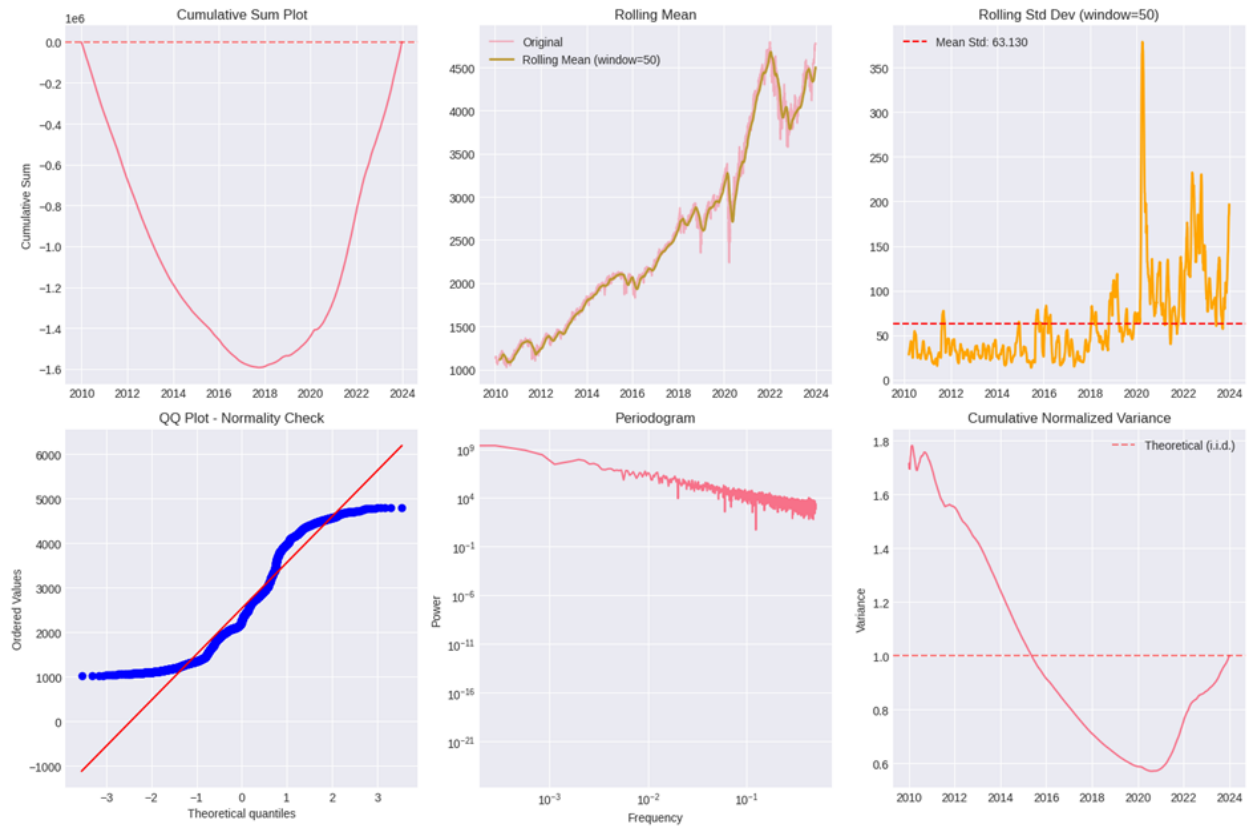
Our simulation findings show that unit root correctly fails to reject the null hypothesis of a unit root. ADF test identifies non stationary, however, the test statistic is negative as but not sufficiently negative to reject the null. We compared real data with simulated AR(1) processes, real S&P 500 prices and returns, simulated unit root, simulated explosive, and simulated stationary. (Brooks 308).

Comparison of AR(1) Processes with Different Roots



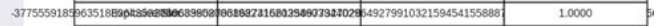
5.9 Unit Root Testing

The S&P 500 closing prices consistently fail to reject the unit root hypothesis across all three ADF test specifications, confirming their non-stationary nature. Our results remain statistically insignificant, showing that the price series exhibits random walk behavior with a drift, where price changes are unpredictable and past trends are not reliable for forecasting (Campbell et al. 22).



5.10 Real Equity Data vs Simulated Processes

The comparison between real equity data and simulated processes goes through different types of autoregressive processes consisting of stationary, unit root and explosive. Stationary shocks have a temporary effect, variance remains constant as show in the chart, explosive processes show variance growing exponentially and unbounded trajectory (Hamilton 45-47), as visualized in Figure.



b) Define the Interaction Variable (Z_t)

Create an interaction term by multiplying the original independent variable $X(t)$ by the dummy variable D_t :

$$Z(t) = X(t) \cdot D_t$$

The variable Z_t will be:

$$Z(t) = \begin{cases} X(t) \cdot 0 = 0 & \text{for } t = 1, 2, \dots, 10 \\ X(t) \cdot 1 = X(t) & \text{for } t = 11, \dots, 20 \end{cases}$$

c) Construct the Unified Regression Model

The single, unified regression model incorporating the structural break in the slope is:

$$Y(t) = \alpha + \beta_1 X(t) + \delta Z(t) + \varepsilon(t) \quad \text{for } t = 1, 2, \dots, 20$$

Here, β_1 is the slope for the first period, and δ (delta) is the change in the slope from the first period to the second period.

d) Interpretation of the Unified Model

For the Pre-Break Period:

$t = 1, \dots, 10$: $D_t = 0$ and $Z(t) = 0$. The Model becomes

$$Y(t) = \alpha + \beta_1 X(t) + \delta(0) + \varepsilon(t) = \alpha + \beta_1 X(t) + \varepsilon(t)$$

This matches the first period's specification with slope β_1

For the Post-Break Period:

$t = 11, \dots, 20$: $D_t = 1$ and $Z(t) = X(t)$. The Model becomes

$$Y(t) = \alpha + \beta_1 X(t) + \delta X(t) + \varepsilon(t) = \alpha + (\beta_1 + \delta) X(t) + \varepsilon(t)$$

This matches the second period's specification with slope β_2 , where $\beta_2 = \beta_1 + \delta$

e) The Hypothesis Test

The question is whether there is a structural break in the slope, which means testing if the slope in the second period β_2 is different from the slope in the first period β_1 .

Since $\beta_2 = \beta_1 + \delta$, the test for a structural break in the slope is equivalent to testing whether the parameter δ is significantly different from zero.

Null Hypothesis (H_0 -No Structural Break): The slope coefficient is different in the second period.

$$H_0: \delta = 0 \quad (\Rightarrow \beta_1 = \beta_2)$$

Alternative Hypothesis (H_A -Structural Break Exists): The slope coefficient is different in the second period.

$$H_A: \delta \neq 0 \quad (\Rightarrow \beta_1 \neq \beta_2)$$

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